

Life Cycle Analysis of Classical Grand and Electric Pianos

Claudia Nanez
Professor Gordon Krauss
Sustainable Design Spring 2024

Background

Since the invention of the piano in the 18th century, the percussive string instrument has been a staple in almost every genre of music. Pianos have had a long history of questionable sustainability, as up until the mid 20th century most piano manufacturers used ivory from elephant tusks for the keys.¹ Steinway & Sons, the American piano manufacturer that has produced world renowned hand crafted pianos since the 1800s, made the switch in the 1950s. In recent years, Steinway has become increasingly committed to improving their sustainability efforts, from working with only FSC-Certified Lumber Suppliers to utilizing solar power in their manufacturing plant.² However, the rigorous raw material acquisition process and the extreme transportation methods involved in the construction of a grand piano results in high levels of emission and waste. Today, many musicians have opted to use an electronic keyboard instead, as they are cheaper and easier to carry. The brand Nord creates high quality keyboards with weighted keys that are designed to sound and feel like a real piano.³ Yet, they are made primarily out of plastic and require electricity. Thus, this life cycle analysis (LCA) aims to determine if electric pianos or classic pianos are more sustainable, and to look holistically at why an individual may choose one option over the other.

Methods

SimaPro LCA software was used to perform the analysis. The assumptions are below:

- Generalizing the type of wood used for the classical pianos
- Making assumptions about the plastic composition of the keyboard and keyboard accessories, such as pedals, connection cables, and stand
- Approximating the energy values for both the classical and electric piano creation
- Assuming waste disposal processes and lifecycle time, while understanding that in reality, this varies case by case



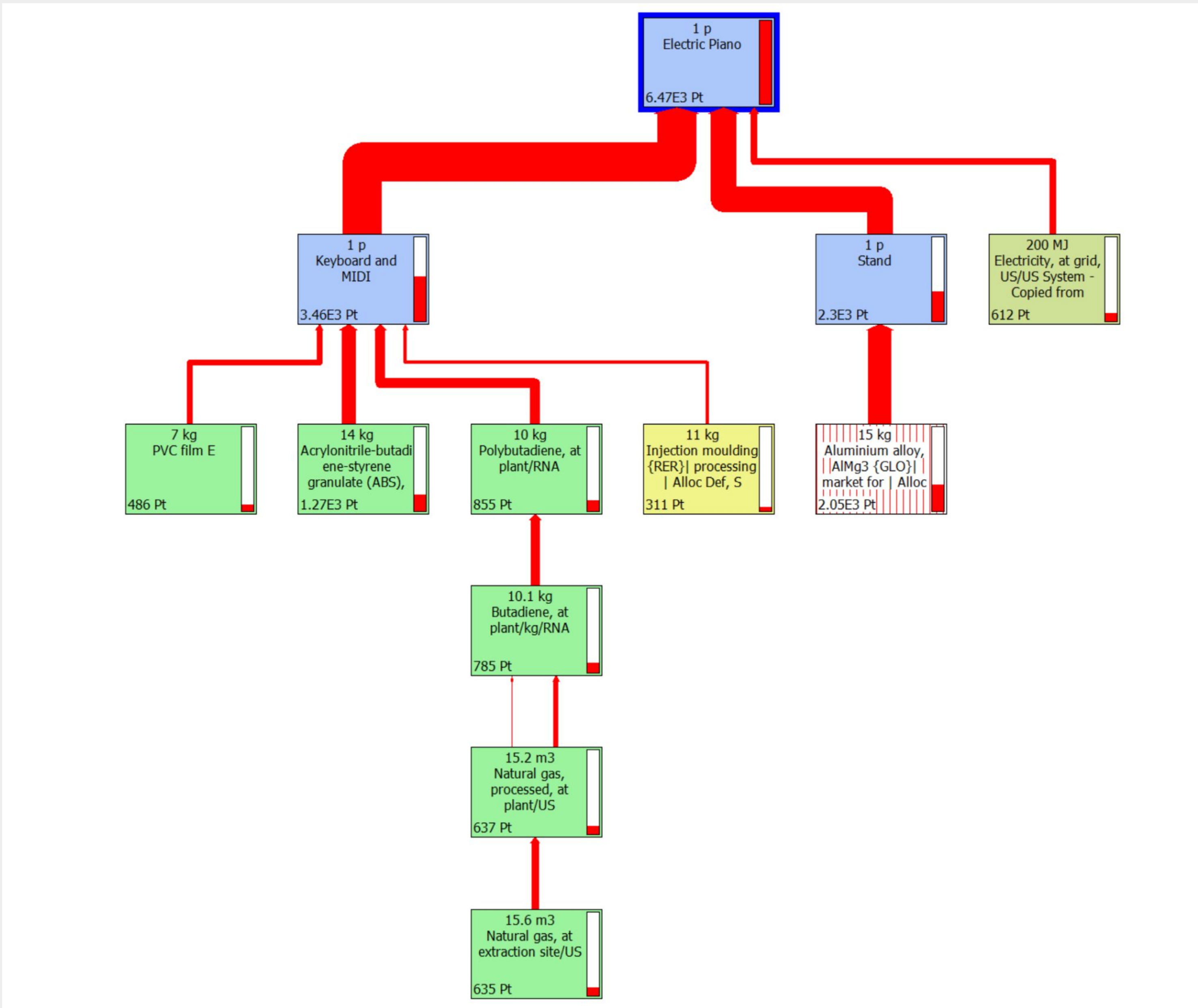
Steinway & Sons Model B Grand Piano



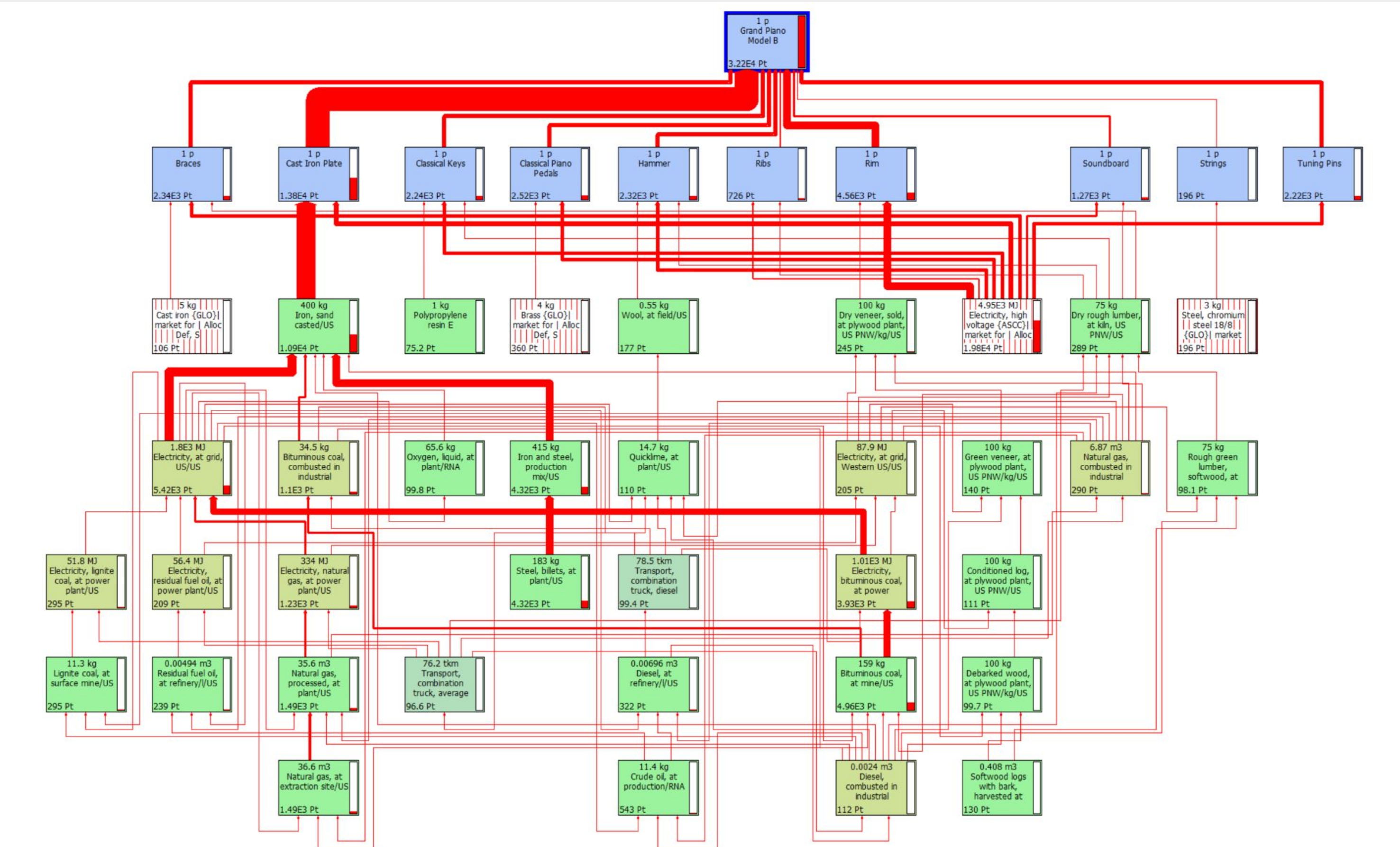
Nord Grand 2 Keyboard

Case 1: Household Use

The household use case assumes that the instrument will be brought to a location and will remain for at least a human lifetime.



Electric Piano Manufacturing and Use Tree



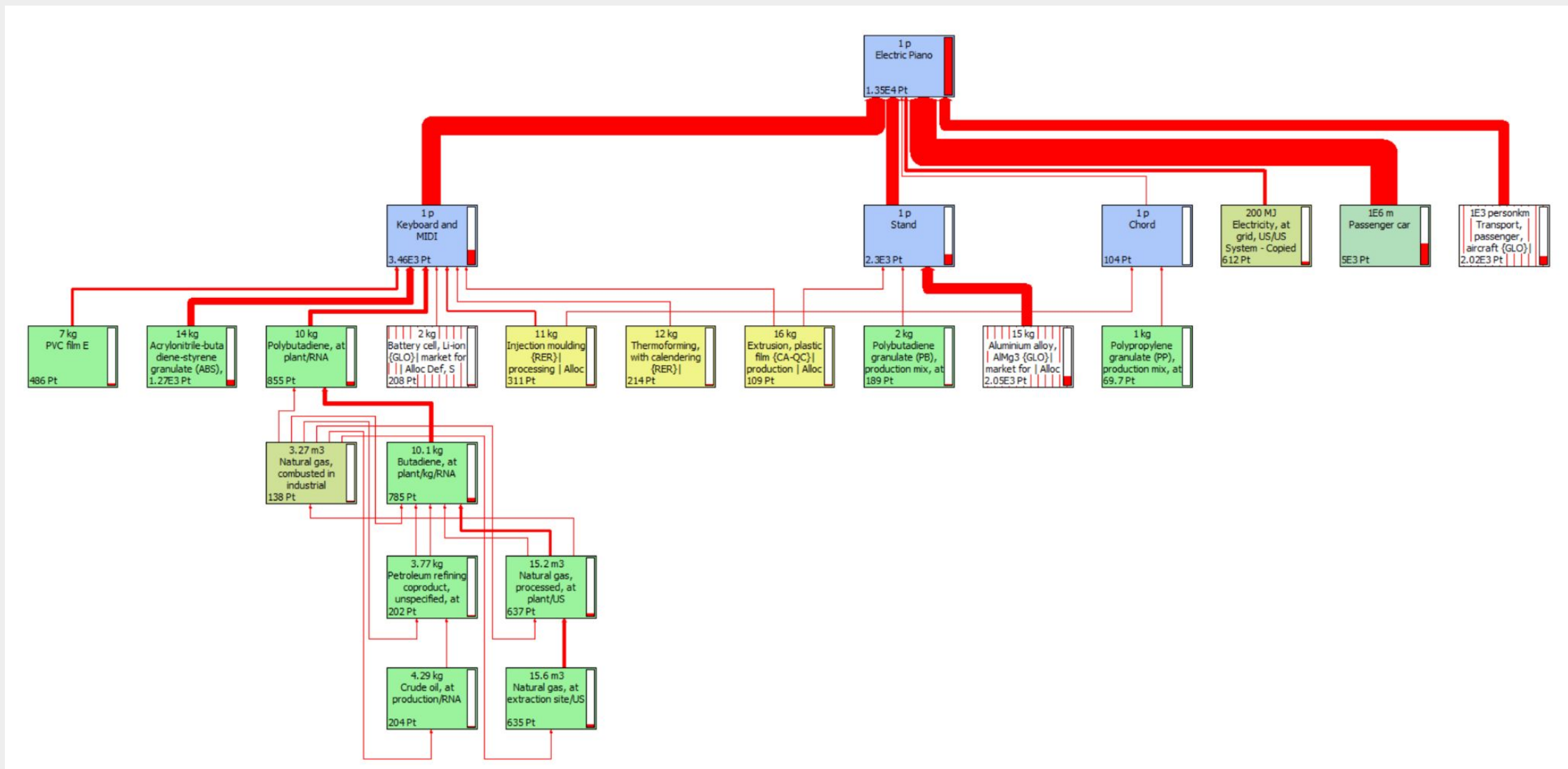
Classic Grand Piano Manufacturing and Use Tree

Discussion

The raw material acquisition, manufacturing, and transportation process of a classical piano is a far more involved process than that of an electric piano. For example, the classic grand piano requires various raw materials such as wool, wood, and different types of metals. Most notably, the cast iron plate of the classic piano contributes to the largest form of emissions. However, large levels of plastic and aluminum involved in the construction of the electric piano are non-renewable. For household use, the choice of piano can be up to the musicians discretion. With traveling, the emissions generated from the truck for the classic piano are larger than that of the passenger car. If the musician travels exclusively by plane, then the additional truck emissions will result in high levels of emissions for the whole tour. Most musicians should use an electric piano when traveling.

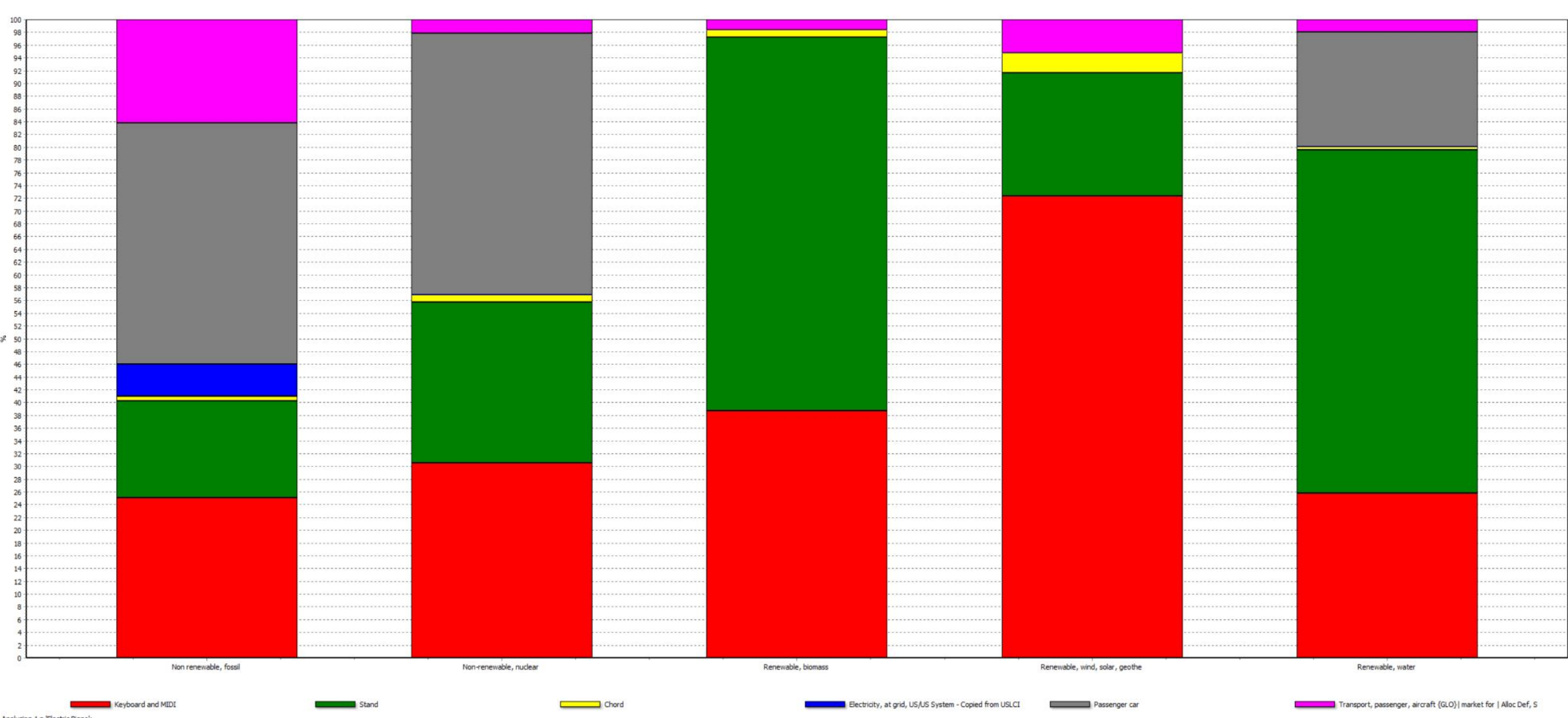
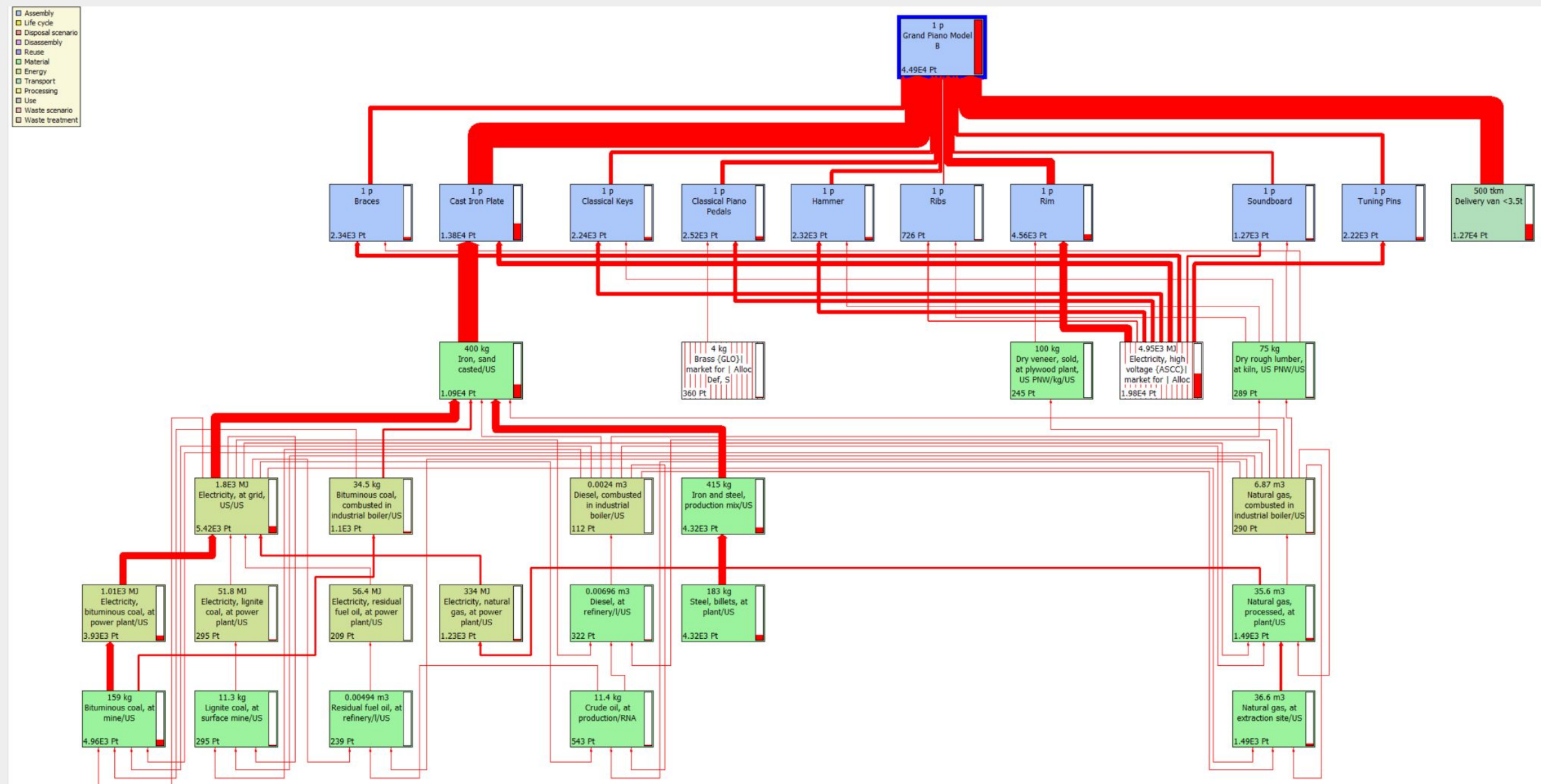
Case 2: Professional Musician

The professional musician case assumes that the musician completes a small tour of about 2,000km .



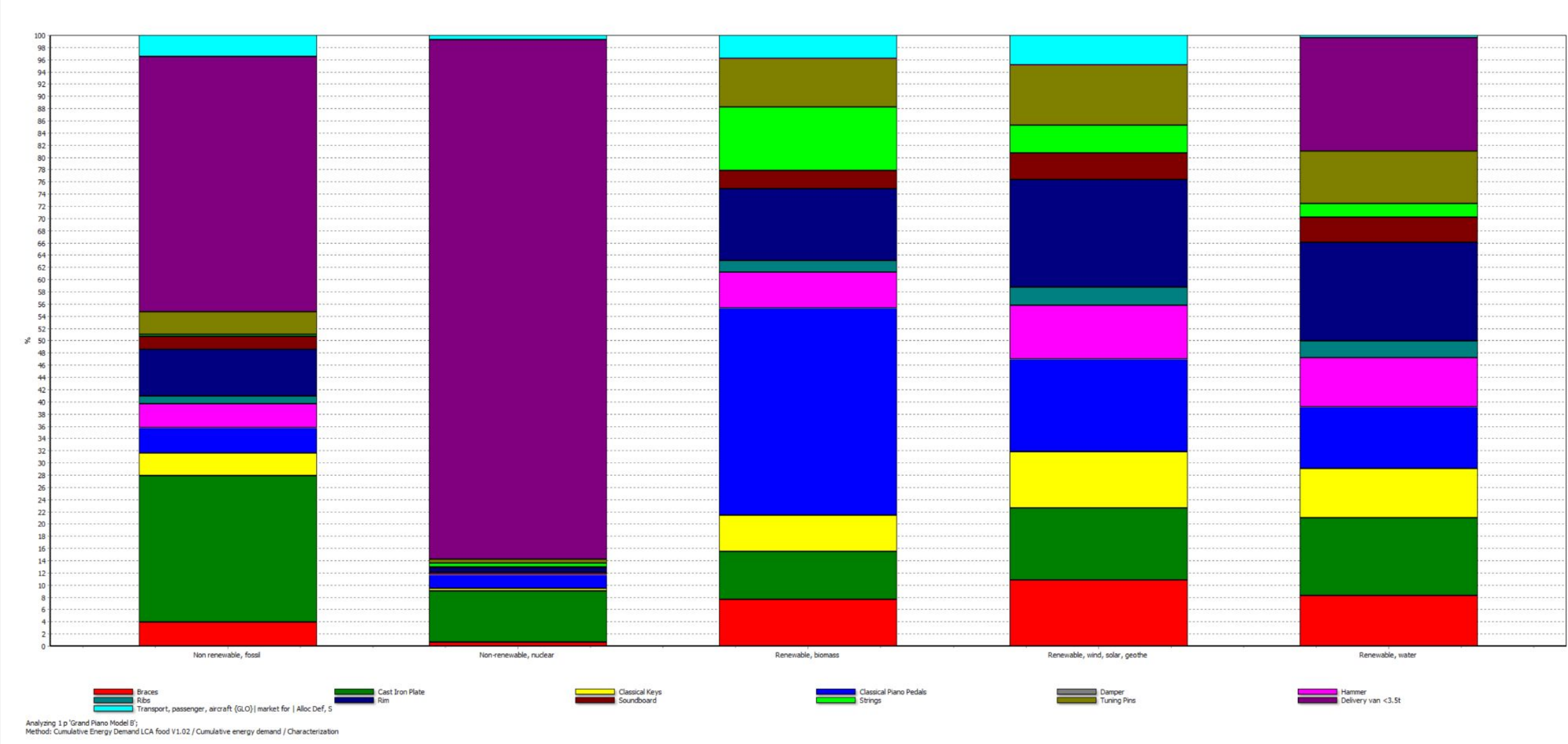
Electric Piano Tree; the piano travels by both air and passenger car with the musician.

Classic Piano Tree; the piano travels exclusively by truck.



Electric Piano Impact Assessment

Classic Grand Piano Impact Assessment



Future Analysis

A large limitation to this analysis has been the lack of specific materials offered in the SimaPro libraries available. Therefore, future analysis to further expand on the conclusions would need to include the specific types of woods, felt, wool, and plastics used for the classical grand piano. Similarly, the plastic types and manufacturing processes involved for the Nord Grand 2 were not known for the analysis.

Sources

¹ Iglauer and Gill, "Change of Keys," The New Yorker, January 1958, <https://www.newyorker.com/magazine/1958/01/25/change-of-keys>

² "Sustainability," Steinway and Sons, <https://www.steinway.com/about/social-responsibility/sustainability>

³ "Specification," Nord Keyboards, <https://www.nordkeyboards.com/products/nord-grand-2/specification>